

Week 01 · What generative AI does

OPEN / WEEK 01

The late-bus refund

A confident assistant says a late bus ticket is reimbursed. Policy P1 says student travel is not reimbursed. Would you trust the answer?



First vote: what would you do?

QUESTION / WEEK 01

A confident assistant says a late bus ticket is reimbursed. Policy P1 says student travel is not reimbursed. Would you trust the answer?

Choose an initial answer and name one piece of evidence you need.

VOTE FIRST • GIVE A REASON

A Accept the first plausible result

B Check the source or measurement

Which source or test could settle this?

The journey through today's case

OPEN / WEEK 01

We will predict, connect an analogy, trace a mechanism, work an example, then explain a decision.

1

Generation and prediction

2

Model and system

3

Prompt and context

4

Baseline and variability

1. Predict · Generation and prediction

QUESTION / WEEK 01

Can a newly worded answer still be false?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A Yes. Novel wording does not establish factual support.

B Every generated sentence is creative evidence.

Which source or test could settle this?

Generation and prediction · a familiar picture

ANALOGY / WEEK 01

A weather forecast predicts a category; a novelist writes a fresh sentence.

What matches our case, and where does the analogy stop?

A weather forecast predicts a category; a novelist writes a fresh sentence.



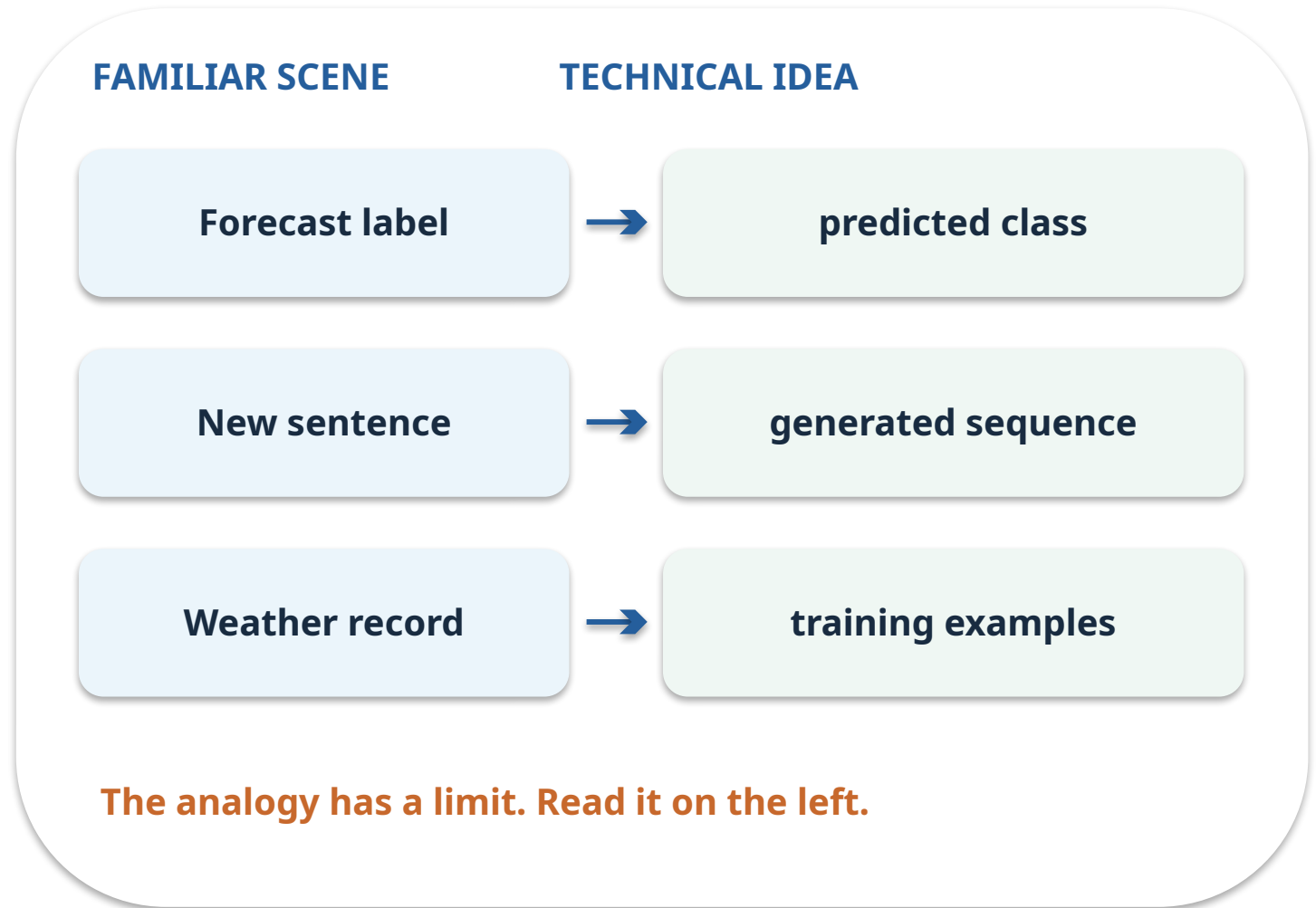
What matches the technical process?

Generation and prediction · map the parts

MAPPING / WEEK 01

Match each everyday part to the technical part.

Analogy limit: A generative model also predicts the next token internally; the distinction concerns the output task.



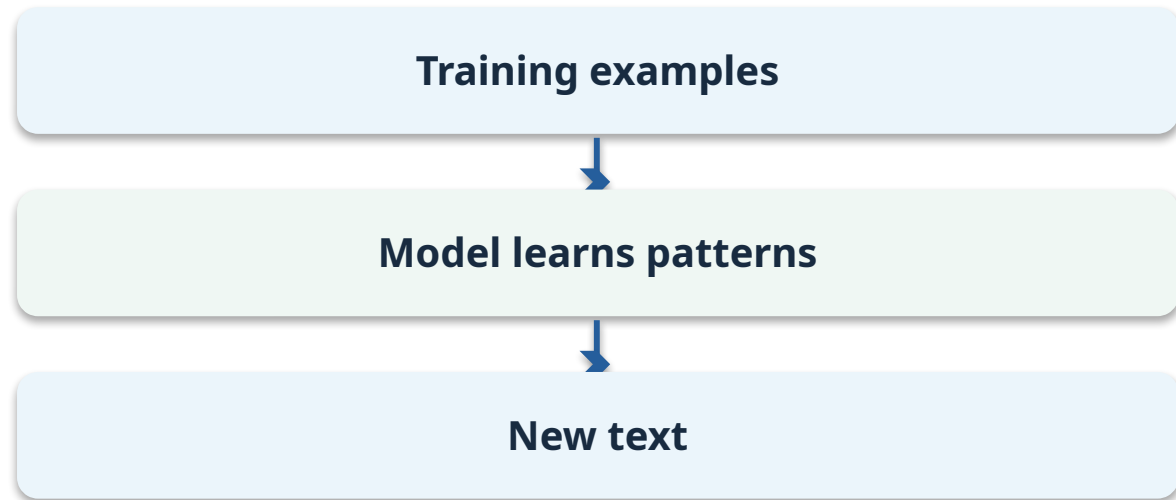
Generation and prediction · how it works

MECHANISM / WEEK 01

A predictor assigns a label or value. A generator creates a new sequence or image. Both can make errors.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW



Generation and prediction · work one case

WORKED / WEEK 01

Given “Is travel reimbursed?”, a classifier could output NO; a generator could write “No, under P1.”

Point to the evidence before giving your conclusion.

GIVEN

Given “Is travel reimbursed?”, a classifier could output NO; a generator could write “No, under P1.”

QUESTION

Can a newly worded answer still be false?

SUPPORTED ANSWER

Yes. Novel wording does not establish factual support.

Generation and prediction · answer and explain

REVEAL / WEEK 01

Yes. Novel wording does not establish factual support.

Tempting mistake: Every generated sentence is creative evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Every generated sentence is creative evidence.

SUPPORTED EXPLANATION

Yes. Novel wording does not establish factual support.

A generative model also predicts the next token internally; the distinction concerns the output tas...

2. Predict · Model and system

QUESTION / WEEK 01

Which component can stop an unsupported answer reaching the student?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A The application check using P1 can flag or block it.

B The model itself guarantees the college rule.

Which source or test could settle this?

Model and system · a familiar picture

ANALOGY / WEEK 01

A cook proposes a dish; the restaurant checks ingredients and allergies before serving.

What matches our case, and where does the analogy stop?

A cook proposes a dish; the restaurant checks ingredients and allergies before serving.



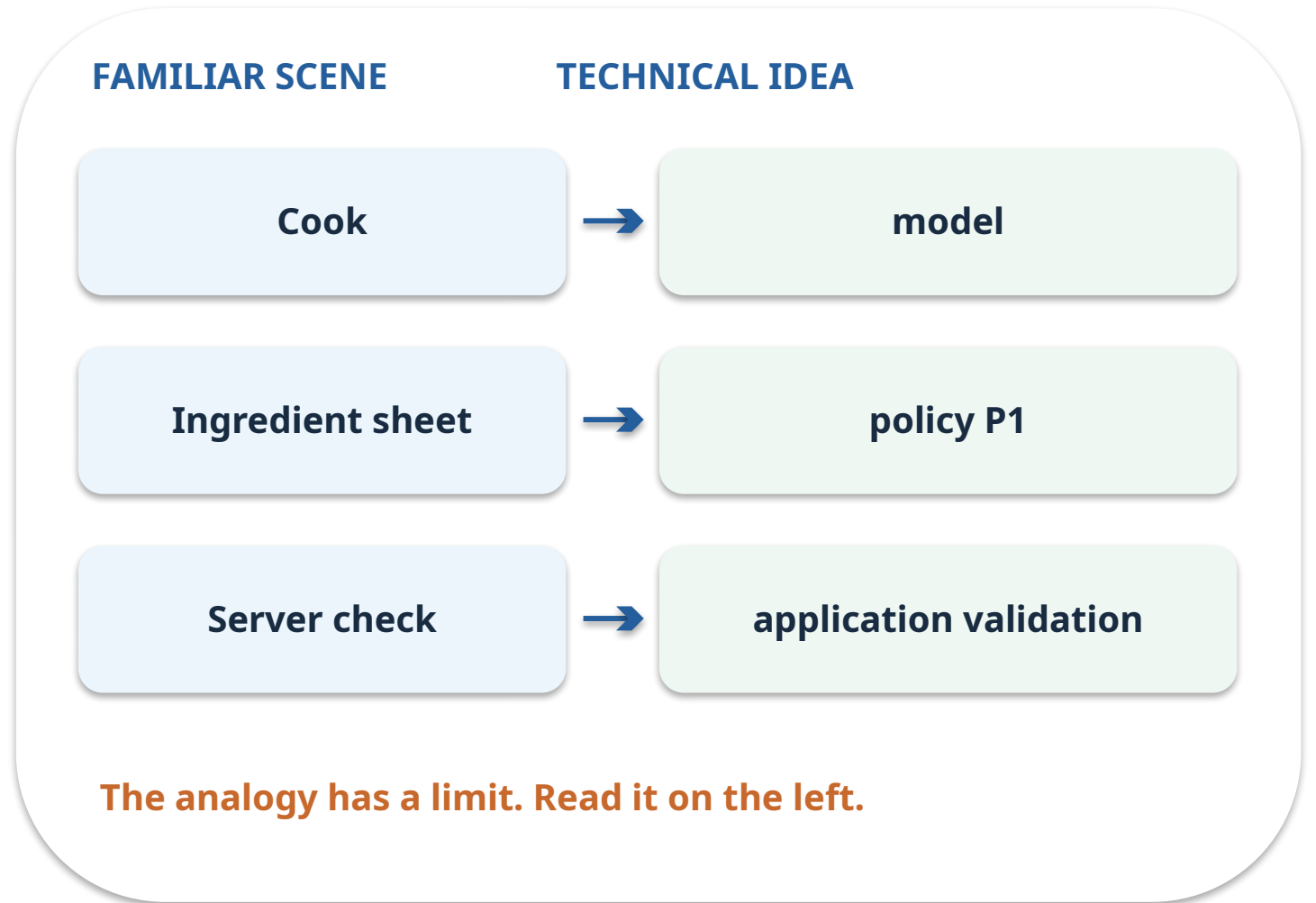
What matches the technical process?

Model and system · map the parts

MAPPING / WEEK 01

Match each everyday part to the technical part.

Analogy limit: A restaurant check is a deliberate rule; a model does not automatically enforce permissions.



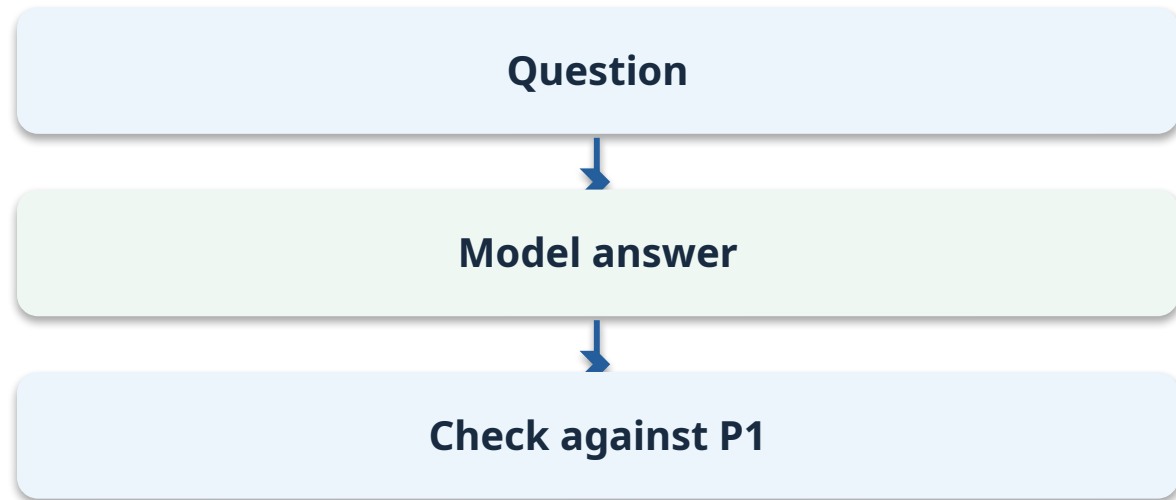
Model and system · how it works

MECHANISM / WEEK 01

The model proposes words.
Retrieval, permissions,
validation and people decide
what the application may
present.

Trace the arrows from left to
right.

FOLLOW THE DATA THROUGH EACH ARROW



Model and system · work one case

WORKED / WEEK 01

The model proposes “Yes”; the application compares the claim with P1 and flags it.

Point to the evidence before giving your conclusion.

GIVEN

The model proposes “Yes”; the application compares the claim with P1 and flags it.

QUESTION

Which component can stop an unsupported answer reaching the student?

SUPPORTED ANSWER

The application check using P1 can flag or block it.

Model and system · answer and explain

REVEAL / WEEK 01

The application check using P1 can flag or block it.

Tempting mistake: The model itself guarantees the college rule.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

The model itself guarantees the college rule.

SUPPORTED EXPLANATION

The application check using P1 can flag or block it.

A restaurant check is a deliberate rule; a model does not automatically enforce permissions.

3. Predict · Prompt and context

QUESTION / WEEK 01

What should the prompt say when P1 lacks the answer?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A State that the assistant should say evidence is missing.

B Longer prompts always create true answers.

Which source or test could settle this?

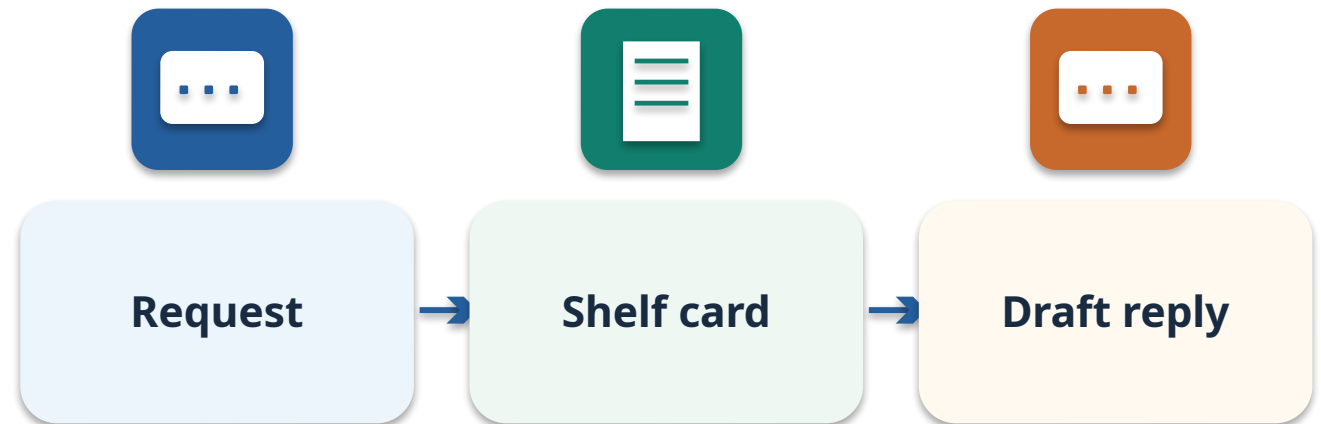
Prompt and context · a familiar picture

ANALOGY / WEEK 01

A librarian needs both the question and the correct shelf card.

What matches our case, and where does the analogy stop?

A librarian needs both the question and the correct shelf card.



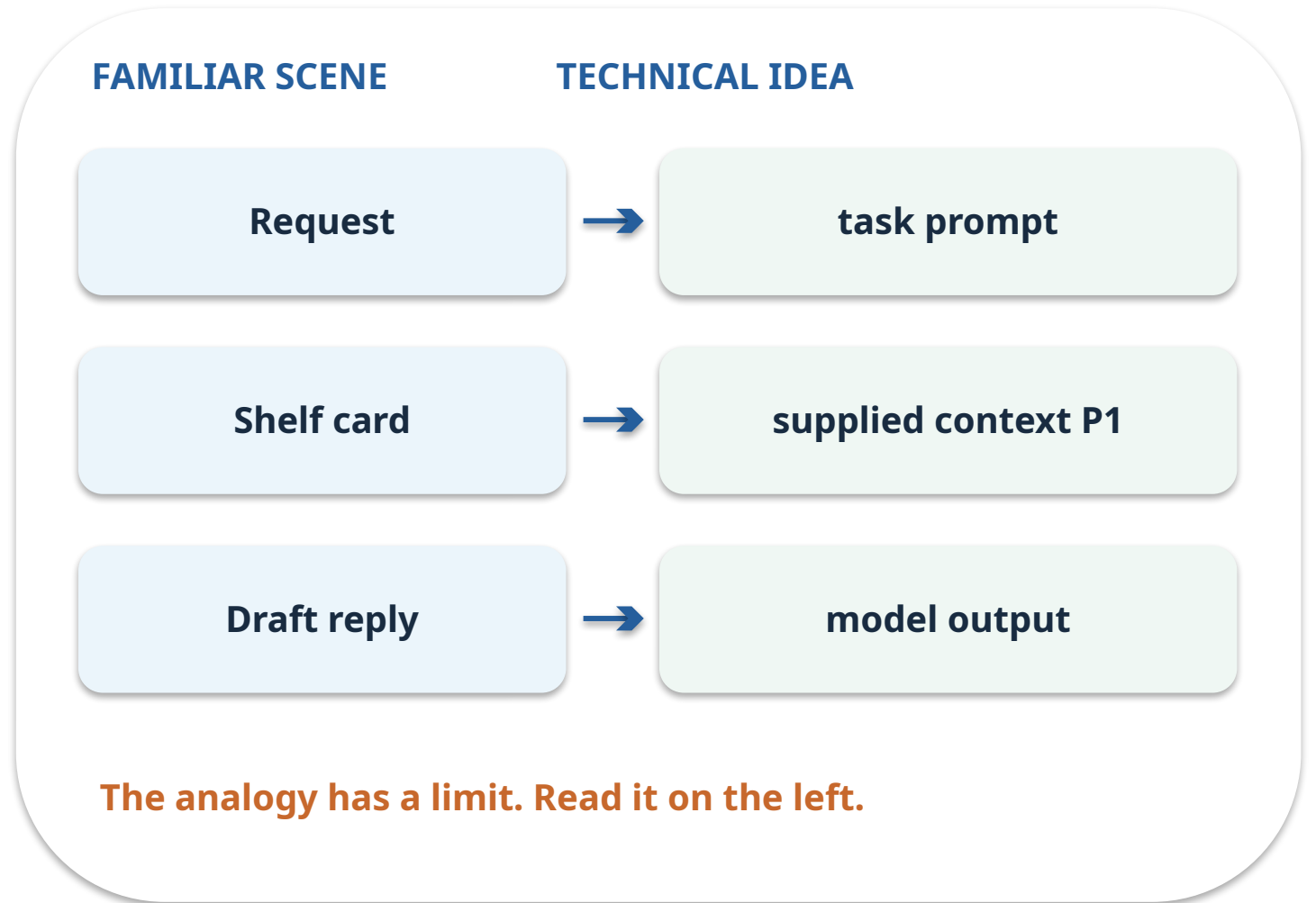
What matches the technical process?

Prompt and context · map the parts

MAPPING / WEEK 01

Match each everyday part to the technical part.

Analogy limit: The shelf card helps only if it is relevant, current and actually used.



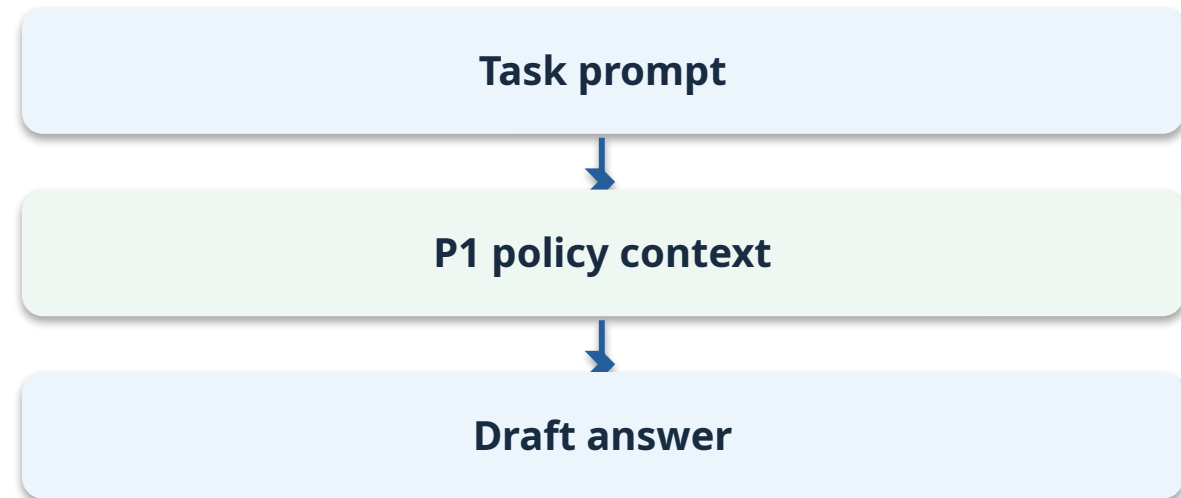
Prompt and context · how it works

MECHANISM / WEEK 01

A prompt states the task;
context contains the supplied
evidence. The model cannot
reliably invent a missing policy.

Trace the arrows from left to
right.

FOLLOW THE DATA THROUGH EACH ARROW



Prompt and context · work one case

WORKED / WEEK 01

“Answer using P1 and cite it” includes a task and source; “help me” does not.

Point to the evidence before giving your conclusion.

GIVEN

“Answer using P1 and cite it” includes a task and source; “help me” does not.

QUESTION

What should the prompt say when P1 lacks the answer?

SUPPORTED ANSWER

State that the assistant should say evidence is missing.

Prompt and context · answer and explain

REVEAL / WEEK 01

State that the assistant should say evidence is missing.

Tempting mistake: Longer prompts always create true answers.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

Longer prompts always create true answers.

SUPPORTED EXPLANATION

State that the assistant should say evidence is missing.

The shelf card helps only if it is relevant, current and actually used.

4. Predict · Baseline and variability

QUESTION / WEEK 01

Why record the denominator and repeats?

Make a choice before we explain the diagram.

VOTE FIRST · GIVE A REASON

A The denominator shows sample size; repeats show whether outputs change.

B A 70% score means the next answer is certainly correct.

Which source or test could settle this?

Baseline and variability · a familiar picture

ANALOGY / WEEK 01

A teacher marks ten scripts against one answer key, then repeats two questions.

What matches our case, and where does the analogy stop?

A teacher marks ten scripts against one answer key, then repeats two questions.



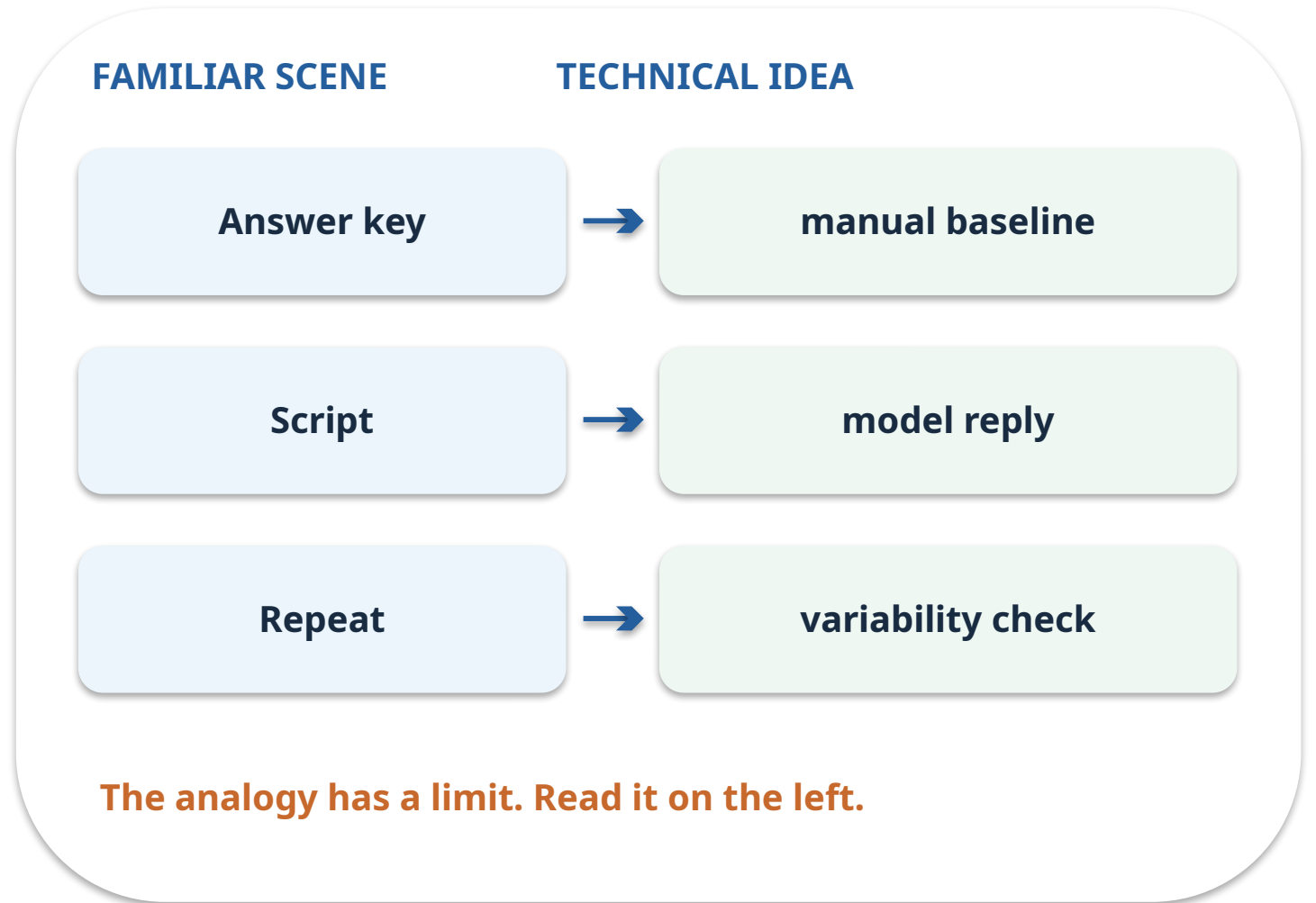
What matches the technical process?

Baseline and variability · map the parts

MAPPING / WEEK 01

Match each everyday part to the technical part.

Analogy limit: One short sample cannot estimate performance in all future situations.



Baseline and variability · how it works

MECHANISM / WEEK 01

A manual answer sheet gives us a reference. Repeated responses can vary, so record both correctness and changes.

Trace the arrows from left to right.

FOLLOW THE DATA THROUGH EACH ARROW

10 reference answers



10 observed answers



Count supported

Baseline and variability · work one case

WORKED / WEEK 01

Seven of ten replies are supported: $7/10 = 70\%$; log the three failures and two repeats.

Point to the evidence before giving your conclusion.

GIVEN

Seven of ten replies are supported: $7/10 = 70\%$; log the three failures and two repeats.

QUESTION

Why record the denominator and repeats?

SUPPORTED ANSWER

The denominator shows sample size; repeats show whether outputs change.

Baseline and variability · answer and explain

REVEAL / WEEK 01

The denominator shows sample size; repeats show whether outputs change.

Tempting mistake: A 70% score means the next answer is certainly correct.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A 70% score means the next answer is certainly correct.

SUPPORTED EXPLANATION

The denominator shows sample size; repeats show whether outputs change.

One short sample cannot estimate performance in all future situations.

Evidence trail: input → result

WORKED / WEEK 01

Q: Is a late bus ticket reimbursed?

A: No; travel is not reimbursed [P1].

GIVEN

Policy P1 says student travel is not reimbursed.

QUESTION

Q: Is a late bus ticket reimbursed?

SUPPORTED ANSWER

A: No; travel is not reimbursed [P1].

Notebook: run the prepared example

PRACTICE / WEEK 01

Run the ten-case baseline and change a single answer.

Read the original output before editing.

1

Read ten short questions and their reference answers

2

Run the supplied rule-based classroom stand-in

3

Mark each answer correct, incorrect or unsupported

4

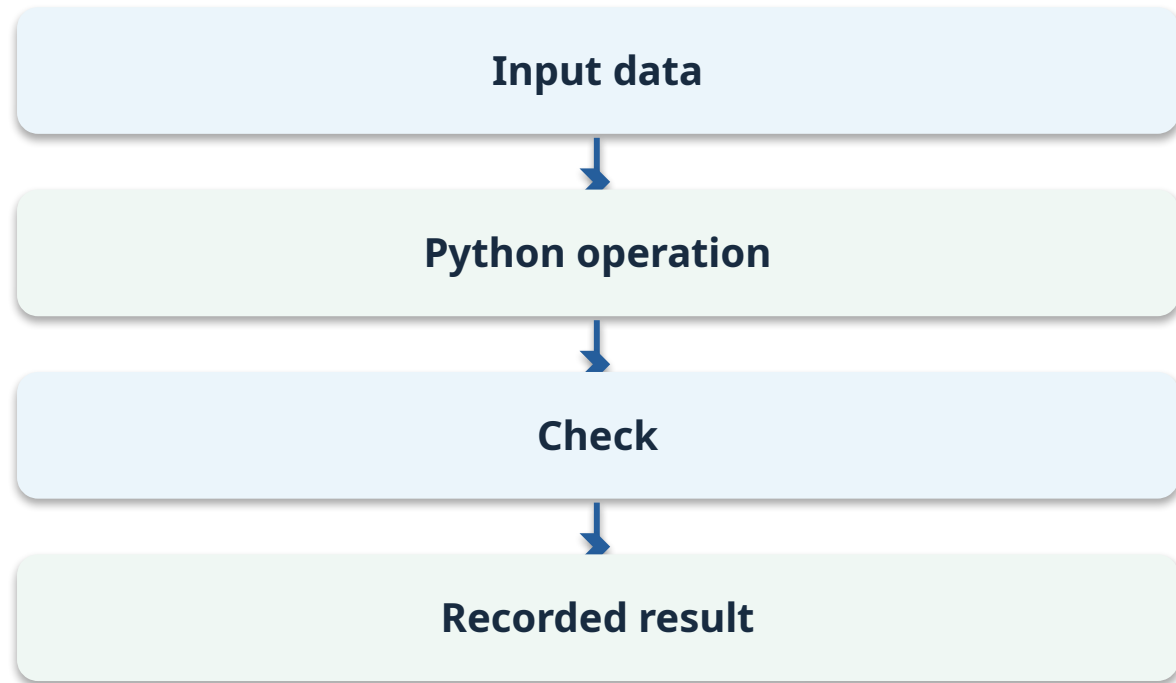
Repeat two questions and compare what changed

Find the input and check in Python

PRACTICE / WEEK 01

Locate the supplied input, the transformation, the checking line and the printed result.

FOLLOW THE DATA THROUGH EACH ARROW



Predict one controlled change

PRACTICE / WEEK 01

Change one answer to an unsupported claim and measure the score again.

What should change in the result? What stays fixed?

Before: original case

After: one change

Keep all other inputs fixed.

Run, compare and record

PRACTICE / WEEK 01

Record the input, the one changed setting, the observed result and a reason for any difference.

Input	Setting	Result
Original	baseline	record
Changed	one edit	compare

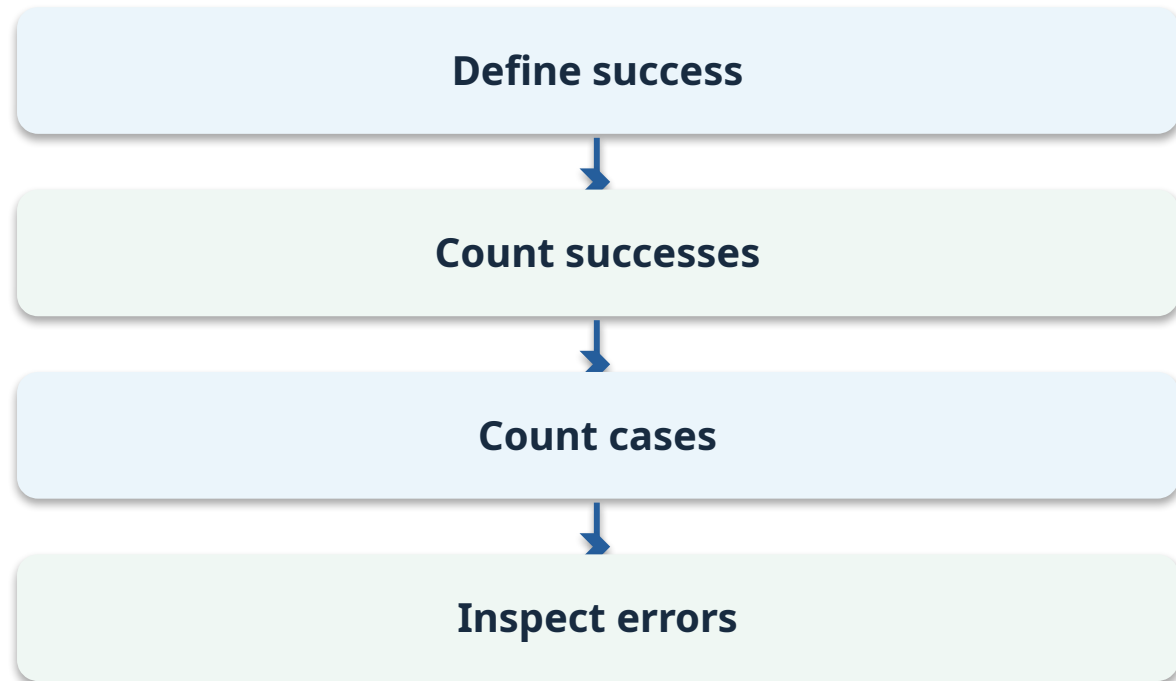
Measure without hiding the denominator

CHECK / WEEK 01

Correct supported answers / 10 questions

Write the numerator, denominator and one failure case.

FOLLOW THE DATA THROUGH EACH ARROW



Error detective: find the layer

CHECK / WEEK 01

Plausible wrong claim: A fluent answer is evidence that it is correct.

Correct it using the case evidence.

TEST THE CLAIM AGAINST THE EVIDENCE

TEMPTING CLAIM

A fluent answer is evidence that it is correct.

SUPPORTED EXPLANATION

Fluency is a property of wording. Check the actual claim against the policy text.

Name a test that would catch this failure.

Transfer to a new situation

CHECK / WEEK 01

The assistant repeats the answer twice, but one version invents an exception. Which answer belongs in the ten-case baseline?

Explain which source or test would settle it.

VOTE FIRST • GIVE A REASON

A Use the visible evidence

B Flag missing or unsafe evidence

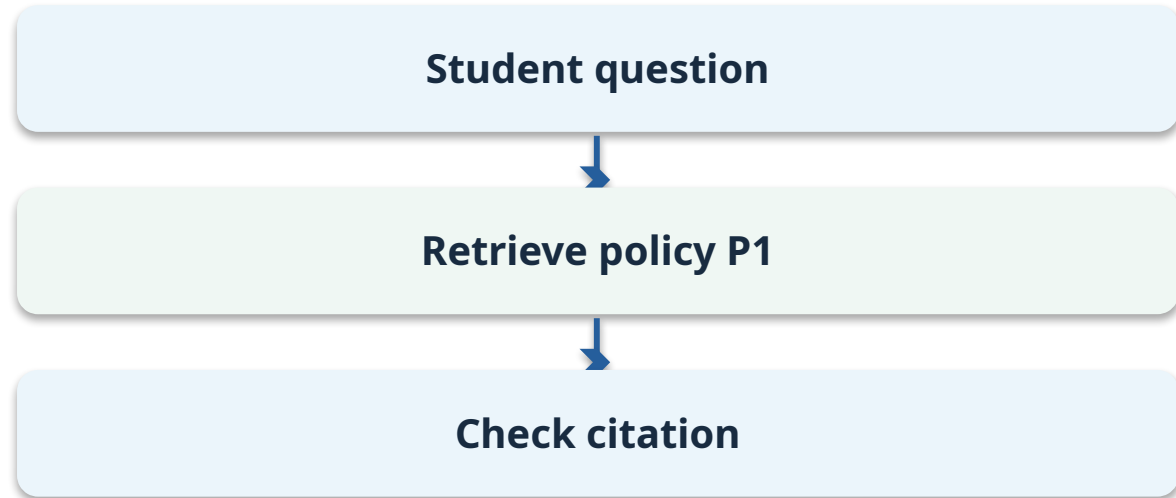
Which source or test could settle this?

Explain it to a partner

CHECK / WEEK 01

Without reading the slide, name the input, one processing step and the check. Your partner should ask “How do you know?”

FOLLOW THE DATA THROUGH EACH ARROW



Exam rehearsal · explain and apply

CHECK / WEEK 01

Explain generation versus prediction with one example of each.

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

Compare with the Word guide

Exam rehearsal · explain and apply

CHECK / WEEK 01

Seven of ten answers are correct and supported. Calculate accuracy and name a missing check.

Write a short answer, then compare with the guide.

1

State the core idea

2

Use a case detail or calculation

3

Add a limitation

4

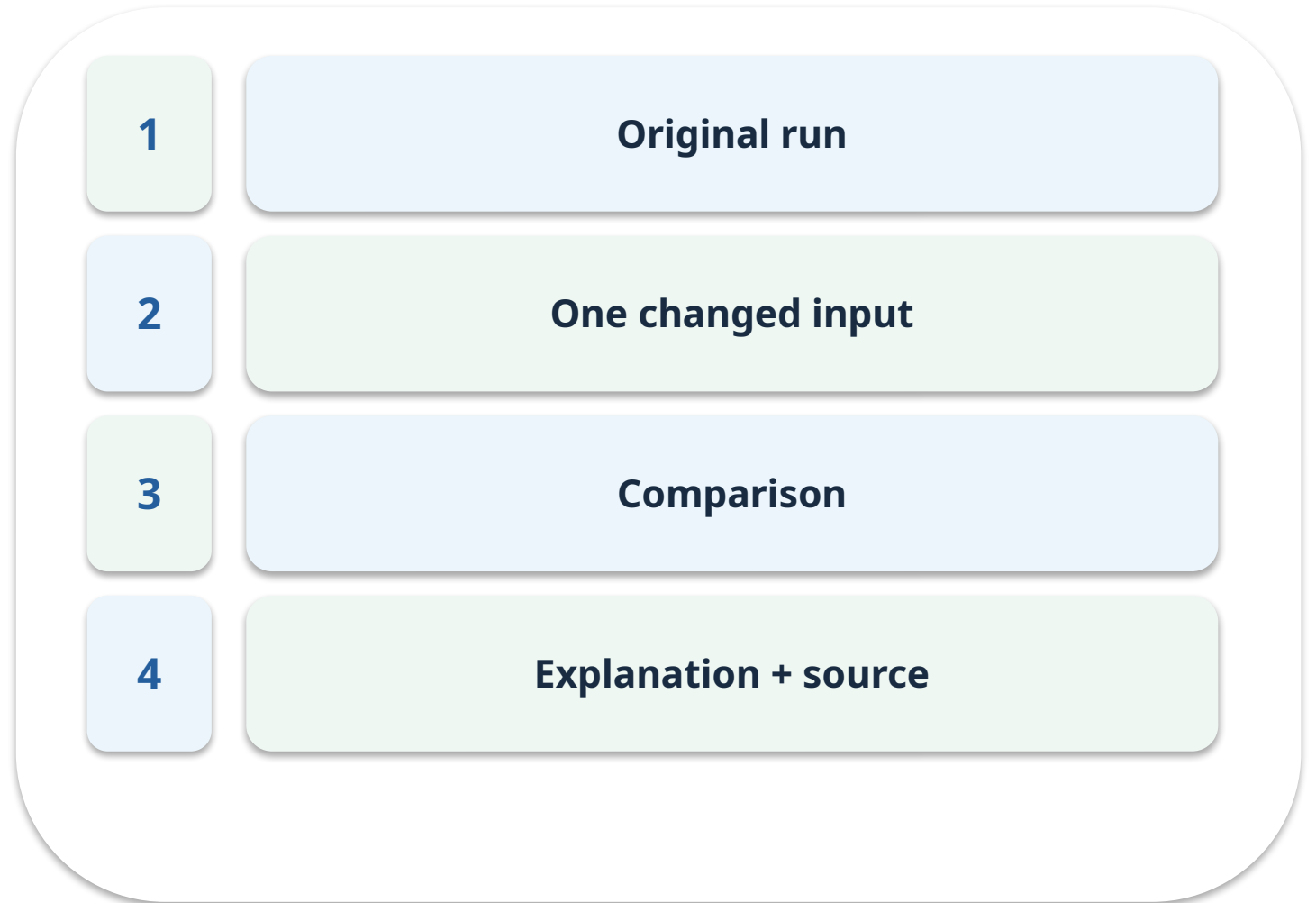
Compare with the Word guide

What to hand in

PRACTICE / WEEK 01

Ten-case answer table with one error analysis.

Include the one changed input and an error explanation.



Three ideas to recall next week

CHECK / WEEK 01

A polished reply is a draft; trace its claim back to P1.

Read the linked official sources in the Word guide and redraw one diagram from memory.

1

Generation and prediction

2

Model and system

3

Prompt and context

4

Baseline and variability